

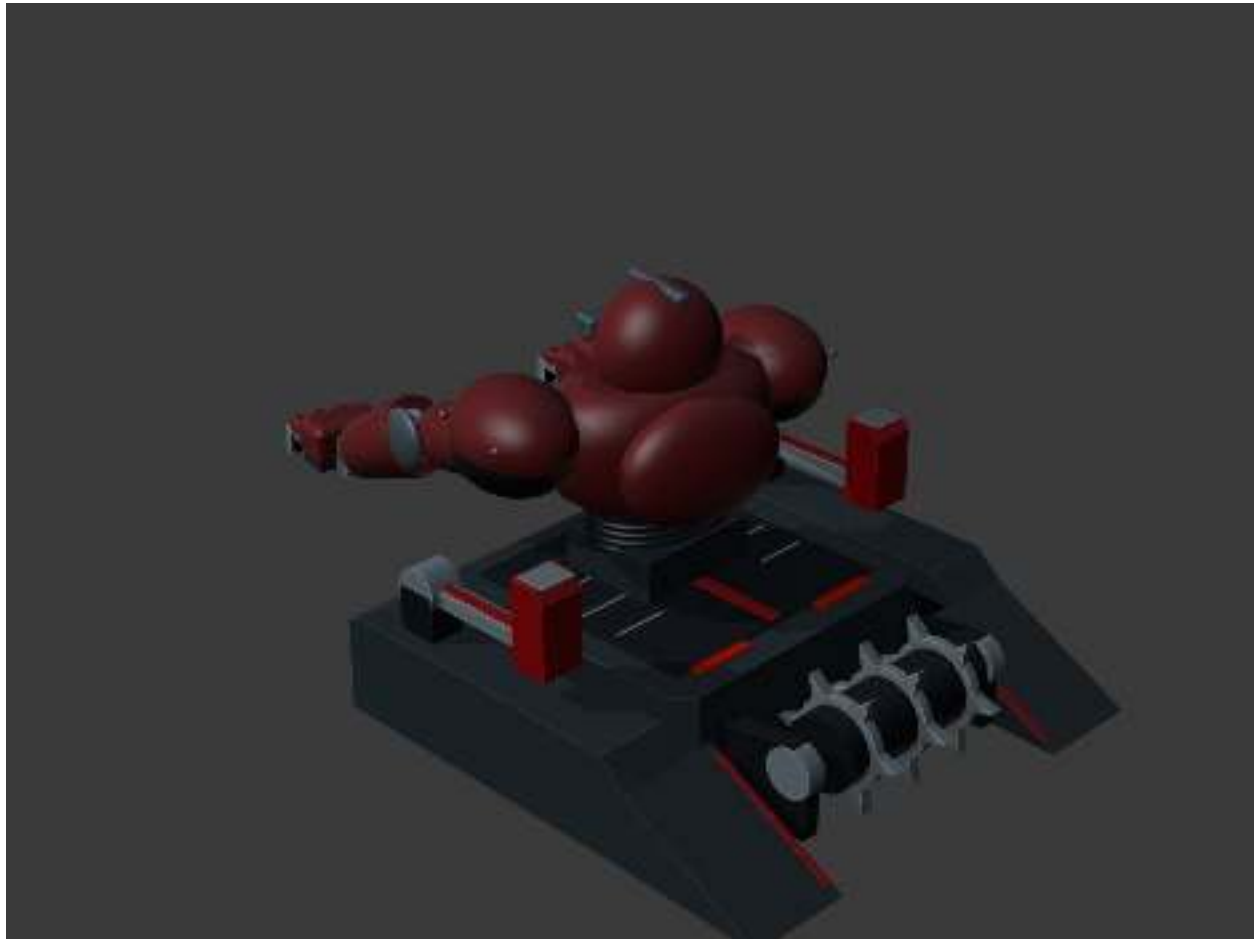
SENTINELT ROBO WARS COMBAT ROBOT

Holistic Build Document

Designer: Thato Glen Assegai

Category: Robo Wars - RC Combat

Competition: Robo Rumble 2026 - Elimination Round



CAD concept | 318.16 mm x 495.00 mm footprint | 180/180 frames PASS

1. Executive Summary

SentinelT is a compact remotely controlled combat-robot CAD concept developed for the Robo Rumble Robo Wars category. The design combines an armored mobile chassis, protected drive layout, sloped external protection, articulated upper structure, serviceable covers, 2.4 GHz RC control architecture, and a guarded active-mechanism provision.

Parameter	Result
Width (X)	318.16 mm
Length (Y)	495.00 mm
Height (Z)	255.41 mm
Maximum allowed footprint	500 mm x 500 mm
Frames checked	180
Failed frames	0
Footprint result	PASS

An engineering mass budget of 4.650 kg was established against the required strictly-below-5 kg limit, leaving a 0.350 kg design allowance. This is a design estimate, not a measured physical mass.

2. Context and Competition Constraints

The elimination challenge is to create a safe, controllable and buildable combat robot while satisfying strict dimensional, mass, control and safety constraints.

Requirement	SentinelT approach
Maximum footprint	500 mm x 500 mm; CAD audited
Mass	Design budget 4.650 kg; physical weighing still required
Control	Human-operated 2.4 GHz RC
Main ON/OFF	Dedicated accessible isolation function
Emergency Stop	Dedicated actuator-power-disable function
Projectiles / flames / liquids	Not used

3. Solution Overview

- Armored mobile chassis
- Protected wheel/drive areas
- Sloped frontal and side guards
- Articulated upper structure
- Removable/serviceable covers
- 2.4 GHz RC architecture
- Dedicated ON/OFF and E-Stop architecture
- Guarded active-mechanism provision

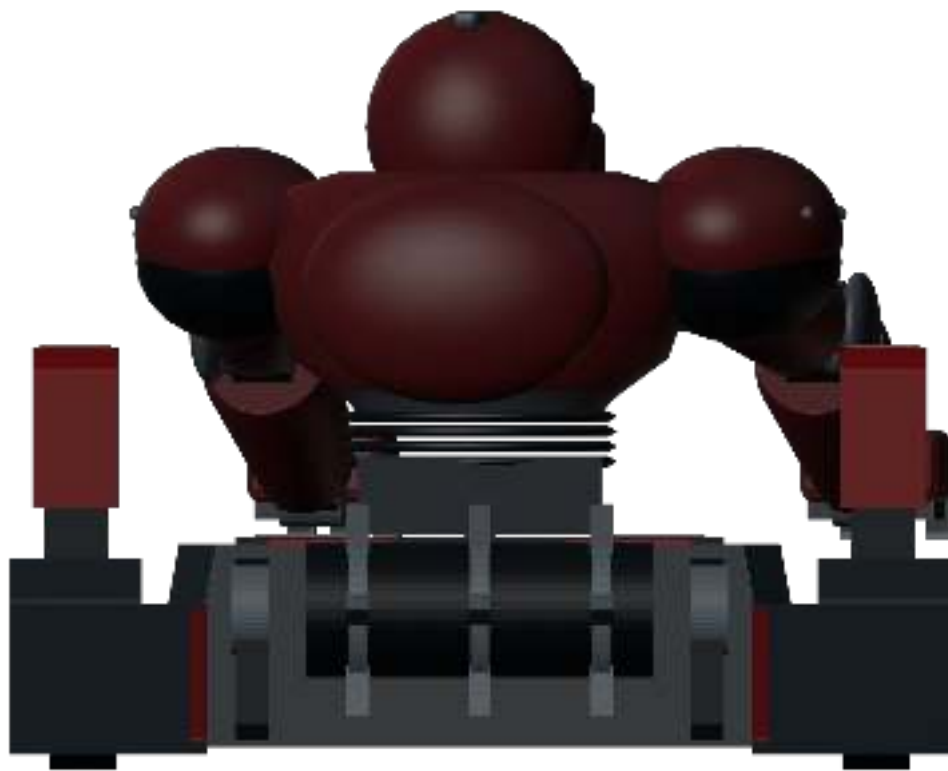


Figure 1. Front CAD view

4. Mechanical Design

4.1 Chassis and protection

The lower chassis is the primary structural and mobility platform. It contains protected wheel areas, sloped guards, a central service volume, an upper service cover, and mounting provision for the upper assembly. The design aims to reduce direct exposure of drivetrain and electronics while remaining compact.

4.2 Upper structure and articulation

The upper structure contains the torso, shoulder housings, articulated arms, forearms, hands and helmet structure. Articulation was deliberately included in the dimensional audit because moving geometry can violate the competition footprint even when a neutral pose is compliant.

4.3 Fabrication approach

The submission is a detailed CAD and engineering concept. Planned fabrication methods include machined or laser-cut structural plate, bent sheet panels, bolted joints, replaceable guards and 3D-printed non-critical covers where appropriate. Final material grades and thicknesses will be selected only after strength, mass and supplier checks.

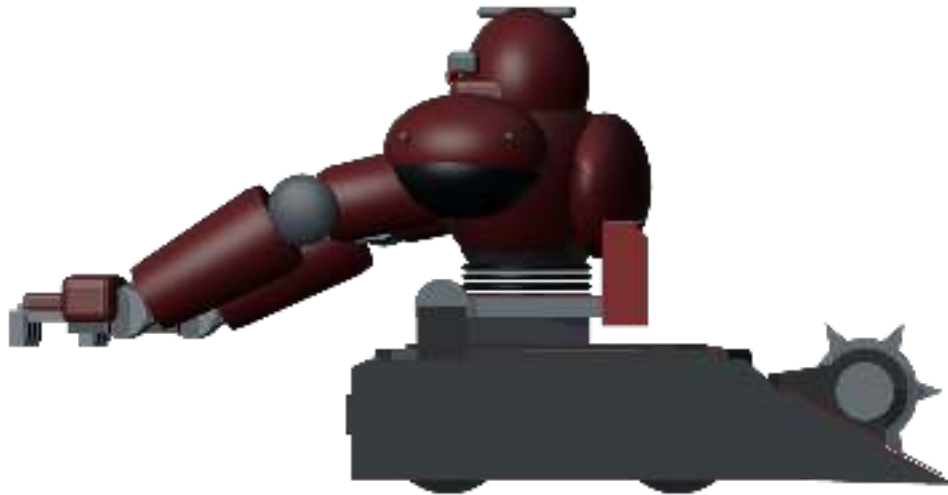


Figure 2. Side CAD view

5. CAD Compliance and Digital Testing

Custom Blender Python scripts were used to calculate world-space mesh bounds and check the design across animation frames 1-180. Earlier geometry exceeded the 500 mm x 500 mm footprint; the final design was repackaged and re-audited until all checked frames complied.

Audit item	Final result
Maximum X	318.16 mm
Maximum Y	495.00 mm
Maximum Z	255.41 mm
Frames checked	180
Failed frames	0
Status	PASS

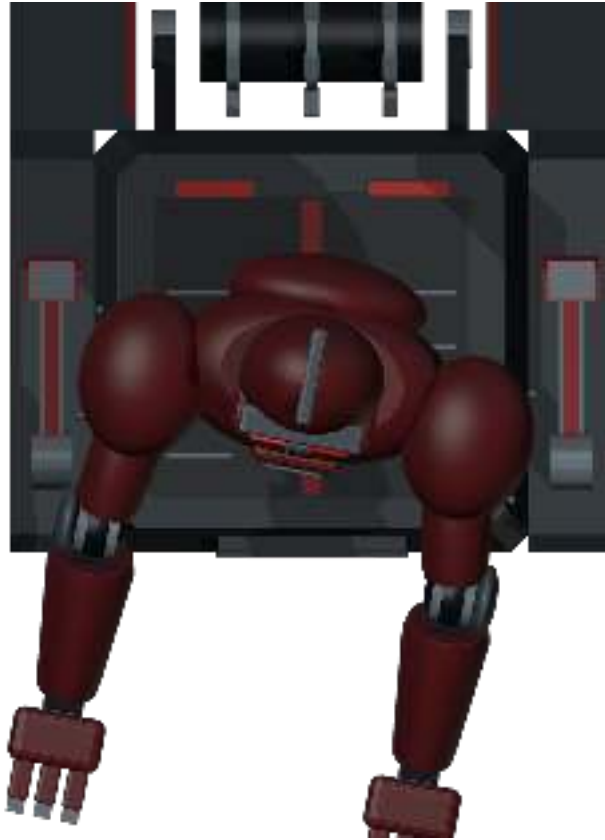


Figure 3. Top CAD view

6. Engineering Mass Budget

Subsystem	Planned mass
Chassis and lower armor	1.10 kg
Drive motors and gearboxes	0.75 kg
Wheels, hubs and shafts	0.30 kg
Battery and power system	0.60 kg
RC electronics, ESCs and wiring	0.25 kg
Guarded active mechanism + drive/mount	0.70 kg
Upper structure, arms and outer shells	0.50 kg
Bearings, brackets and fasteners	0.25 kg
Contingency allowance	0.20 kg
TOTAL	4.65 kg

Remaining design allowance to 5.000 kg: 0.350 kg. The completed robot must be weighed on a calibrated scale before competition; the CAD model alone cannot certify physical mass.

7. Electronic Design and Safety

The proposed power chain is: Battery -> Main Fuse -> MAIN ON/OFF SWITCH -> EMERGENCY STOP / power-disable stage -> Power Distribution -> drive controller, control electronics and auxiliary controller. The 2.4 GHz receiver feeds operator commands to the controller.

7.1 Main ON/OFF switch

The main ON/OFF function provides accessible primary isolation without chassis disassembly. The final switch or disconnect device must be correctly rated for the actual DC load.

7.2 Emergency Stop

The E-Stop is treated as a dedicated hardware safety function intended to disable actuator power immediately. It must not rely only on normal software commands. Exact E-Stop contacts/contact ratings will be selected after battery voltage and peak actuator current are known.

7.3 RC signal safety

Loss of a valid 2.4 GHz RC signal is treated as a safety event. Startup, controller reset, invalid receiver data or signal timeout all return actuator commands to a disabled state.

8. Programming and Framework Design

The software is category-specific: SentinelT is not autonomous. Human commands are read from a 2.4 GHz receiver, transformed into differential-drive outputs, and gated by safety state and auxiliary-enable logic.

- Initialize controller and disable outputs
- Verify E-Stop state
- Initialize/read 2.4 GHz receiver
- Reject invalid or timed-out RC data
- Apply throttle/steering to differential drive
- Keep auxiliary output disabled unless deliberately enabled
- Return to safe state on E-Stop or RC loss

The repository includes a documented flowchart and a hardware-agnostic Arduino/C++ control skeleton. Hardware-specific pin assignments and receiver/ESC calibration will be added only after component selection and bench testing.

9. Bill of Materials and Procurement

The required spreadsheet is stored at Documentation/BOM/SentinelT_Bill_of_Materials.xlsx. It contains component name, quantity, cost per unit, supplier, stock code, direct URL and total cost columns. No parts are currently owned. Unverified supplier data is marked TBD rather than being presented as confirmed.

10. Serviceability and Buildability

The design uses removable covers and modular assemblies so the battery, receiver, motor controllers, wiring and drivetrain can be inspected without dismantling the entire robot. Final fastener types, connector placement and service clearances will be resolved during physical prototyping.

11. Limitations and Required Physical Verification

- Measured final mass

- Final material grades/thicknesses
- Drive torque, speed and current
- Battery runtime and temperature
- RC range and failsafe behaviour
- E-Stop interruption performance
- Structural durability and fastener retention
- Arena performance

These limitations are documented deliberately. They are not claimed as completed physical tests.

12. Individual Role and Responsibilities

Thato Glen Assegaai is the individual designer and is responsible for concept development, CAD modelling, mechanical architecture, Blender scripting and compliance auditing, control-system design, safety architecture, documentation and repository management.

13. Conclusion

SentinelT is a competition-focused Robo Wars CAD concept designed around the elimination-round constraints. The final CAD footprint of 318.16 mm x 495.00 mm complies with the 500 mm x 500 mm limit across 180/180 checked frames. The engineering mass allocation is 4.650 kg, leaving a 0.350 kg design margin below 5 kg, subject to physical weighing. The repository also documents 2.4 GHz RC control, ON/OFF isolation, E-Stop architecture, safe-state control logic, mechanical CAD evidence and simulation/audit workflow. The next phase is final component procurement, electrical rating verification, fabrication and controlled physical testing.